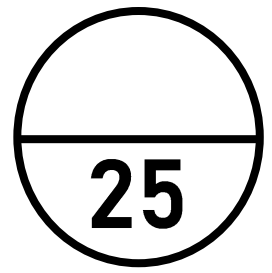


Design Report

Engineering Product Development (EPD) Pillar
30.100 Computational and Data Driven Engineering (CaDDE)
Term 4, 2026



DESIGN REPORT
Hands-On Introduction to Machine Learning for Future Engineers

Due Date: 19th April 2026 (Sunday, 2359 hours)

Type in your answers into the text boxes for each question below after carefully reading through the Design Brief document. This 1D design project is an “**individual**” task, NOT a group work.

File name format for upload should be “**CP_0X_StudentID_1D_Report.pdf**”

Weightage for the 1D Design Project = **25%**

Name:	Chua Heng Li	Task A Final Rank:	
Cohort:	CP02	Task B Final Rank:	
Student ID:	1008931	Task C Final Rank:	

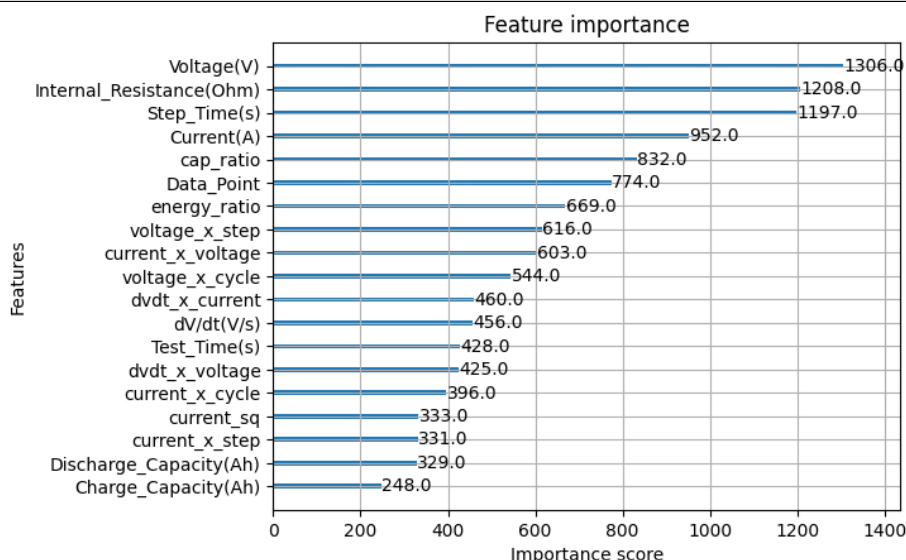
Assessment Component		Your Score	Max Score
A	Kaggle Competition Leaderboard Ranking		50
B	Design Report Reflections		50
A + B	Overall 1D Design Project Score		100
Scaled Score for 1D Design Project			25

FEEDBACK AND COMMENTS TO STUDENT:

Answer the following questions briefly, from your own perspective, following the word limit that is imposed. Note that this section is meant to be a *self-reflection*, so please DO NOT rely on responses from LLMs to answer these questions. Many of these questions do not have a right or wrong answer, we want to hear your views and perspectives.

TASK A. Electrochemical Stress Index (ESI) Prediction

1. Describe one relationship in the dataset that aligns with engineering intuition—and one that surprised you (must include a plot or evidence). (**Word Limit: 100-150 words**)



ANSWER

One relationship that relates to engineering intuition is the high feature importance of Internal Resistance and Current in predicting ESI. From physics perspective, high resistance leads to increased Joule heating, $P = I^2R$, which can accelerate thermal degradation on the battery. This is supported by the Correlation graph showing a large positive linear relationship between these variables.

The surprising relationship is that the combined feature of current_x_voltage is of a lower importance than voltage, current and internal resistance itself. Since engineering intuition is that power = voltage x current which should dominate the importance in creating heat and worsening the battery life. Might be due to the mechanical stress more correlated to the batteries' potential than the power throughout.

2. If you had a detailed knowledge of battery operating and degradation mechanisms as an electrochemical engineer, how would you envision including any physical domain knowledge of battery physics into your ML model? (**Word Limit: 100-150 words**)

ANSWER:

I would integrate physical domain knowledge by using a hybrid approach. First, I would use feature engineering to include non-linear descriptors from Butler-Volmer equation or Nernst equation, ensuring the model recognizes the fundamental relationship between overpotential and reaction rates. Second, I would implement a Physics-Informed Neural Network (PINN) where the loss function includes a regularization term that penalizes predictions violating Kirchhoff's laws or mass conservation. Finally, I could also use residual learning, a electrochemical model would provide a baseline prediction for degradation while ML model would focus on capturing the complex, noisy residuals that traditional physics equations cannot describe. This would incorporate the data model to respect physics constraints for higher accuracy.

3. Did you observe any strong or weak correlation between the various input features and the target? Provide one example and explain how it helped you to simplify your model. (**Word Limit: 100-150 words**)

ANSWER:

I observed a strong positive correlation between Internal Resistance and ESI. From an engineering perspective, higher resistance leads to increased heating and mechanical strain, making it a critical predictor. This strong correlation allowed me to prioritize this feature during model training, ensuring it remained a primary split criterion in my decision trees.

I also identified several weak or non-existent correlations with feature like "Date_Time", "Is_FC_Data", "AC_Impedance(Ohm)". Upon analyzing the data, these were features with no variance and were constant values. By identifying these features, I could simplify my model through removing these redundant columns/features. This reduced the model's dimensionality, decreased training time, prevented the algorithm from assigning importance to noise, and improved generalization on the test set.

4. Describe your worst-performing model. Why do you think it failed? What did you learn from it? (**Word Limit: 100-150 words**)

ANSWER:

My worst performing model was Ridge Regression baseline implemented without feature engineering. This model failed as it assumed a linear relationship between inputs like current, voltage and the target ESI. However in reality electrochemical stress is highly complex and non-linear. As a result, the Ridge model suffered from underfitting, as the linear model could not capture the intricate interactions in the data.

This failure taught me that domain knowledge is important. By just applying a model by itself without knowing anything about the context and underlying mechanics it will be ineffective. This conclusion got me to pivot towards XGBoost, a gradient boosted tree model capable of capturing non-linearities and feature interactions which provided me with a more robust fit for this problem.

5. Did you try modifying the complexity of your model (for example, number of neurons per layer or number of hidden layers in the neural network)? When and how do you decide to stop making the model too complex to avoid overfitting problems? Did you experience overfitting? How did you detect it? (**Word Limit: 100-150 words**)

ANSWER:

During the tuning of XGBoost, I experimented with the complexity by varying max_depth between 3 and 8 and n_estimators up to 1000. I determined that a max_depth of 6 provided the optimal balance and increasing it further led to overfitting, which I detected through 5-fold cross-validation.

I identified overfitting when the training R^2 remained near perfect while the validation (OOV) scores began to fluctuate significantly, indicating high variance and poor generalization. To prevent the model from being too complex, I monitored the gap between training and validation performance. I decided to stop increasing complexity once the validation metrics deteriorated. This systematic approach ensured that the model captured the underlying electrochemical patterns without memorizing the data.

6. If you had more time, what is the first thing you would try to improve your model—and why? (**Word Limit: 100-150 words**)

ANSWER:

Given more time, I would first implement temporal feature engineering and transition to a Recurrent Neural Network (RNN) architecture.

Currently, my XGBoost model treats each data point as independent. However, battery degradation is a sequential process. I could create rolling window statistics like moving averages and standard deviations to capture the historical trends of degradation.

I could also implement a Long Short-Term Memory (LSTM) network. Unlike standard regression models, LSTMs are specially designed to capture long-term dependencies in time series data, making them ideal for modeling how internal resistance evolve over many cycles.

These changes will provide a more accurate representation of the accumulated mechanical and chemical stresses that lead to battery degradation.

TASK B. Engine Pump Anomaly Detection

7. If your model is deployed in a real engineering system, what type of error is most dangerous - False Positive (FP) or False Negative (FN)? Justify using engineering reasoning. (**Word Limit: 100-150 words**)

ANSWER:

If deployed in a real-world engineering system, False Negatives are significantly more dangerous than False Positives. A False Negative occurs when the model classifies critical engine pump failure as normal operating condition, allowing the system to continue to operate under compromised conditions. This can lead to catastrophic mechanical failure, hazardous fluid leaks, and irreversible damage to downstream safety components and could endanger human lives.

On the other hand, False Positives are just misidentifying normal operating conditions as anomaly, which only leads to unnecessary maintenance inspections. While False Positives can lead to inefficiency and financial costs, they do not compromise the physical integrity of the system. Hence model must be tuned to prioritize high recall, ensuring the system captures every possible fault even at the cost of slight expense of precision.

8. How would noisy data or sensor drift affect your model performance? How would you go about “stress testing” your ML algorithm to properly account for these real-world non-idealities? (**Word Limit: 100-150 words**)

ANSWER:

Noisy data or sensor drift can severely degrade performance and causing the model to mistake random fluctuations for anomalies. This results in the data distribution deviates from the training set.

To stress test the model, I would perform sensitivity analysis by injecting noise of increasing variance into the sensor inputs to determine the threshold where the AUC-ROC begins to collapse.

Additionally, I would simulate sensor ageing by applying linear drift offset to specific features over time. To account for these errors, I would implement Data Augmentation during training. Exposing model to noisy samples and using sliding window normalization to adapt to gradual baseline shifts in sensor calibration.

9. How would you use the same autoencoder architecture not just to detect whether the data shows normal / anomaly pattern but also the “type” or “nature” of the anomaly pattern when there are multiple types of anomalies (multi-modal fault signatures) corresponding to different fault signatures in the system being condition monitored? (**Word Limit: 100-150 words**)

ANSWER:

To identify the nature of an anomaly, I would transition from global reconstruction loss to analyzing reconstruction residual vector. Instead of a single value, I would examine which specific sensor dimensions contribute most to error. For example, a pump seal failure might show high residuals in pressure sensors.

I would also use an unsupervised clustering algorithm like K-means or DBSCAN, on these residual vectors or the encoded latent space representations. By mapping different clusters with known historical fault signatures, the autoencoder can transition from binary detection to diagnostic tool. This allows the system to signal that a fault exists and categorize it as a specific failure that is based on a unique “signature” in the error distribution.

TASK C. Battery SEM Degradation Image Classification

10. Explain one hyperparameter you tuned and its effect on performance. Include before / after results. (**Word Limit: 100-150 words**)

ANSWER:

I tuned the learning rate strategy for fine-tuning the model. Initially, applying a uniform learning rate across all layers produced 5-fold cross-validation accuracy of approximately 0.72. I observed that aggressive updates in early layers were destroying the pre-trained weights optimized for edge detection. To improve this, I implemented differential learning rates, using a smaller rate for initial convolutional blocks and larger rate for the classification head. This adjustment improved CV accuracy to 0.88. This strategy allowed the model to preserve low-level feature extractors for SEM textures while allowing classification head to adapt fully to the specific fault signatures of healthy vs fragmented particles. This confirms that weight preservation is important when working with small dataset of 60 images.

11. Describe the role of convolutional filters in your model. What kind of patterns do early vs deep layers capture? How did changing the number of layers or filters affect your results?. (**Word Limit: 100-150 words**)

ANSWER:

In my model, convolutional filters are used as spatial hierarchy for feature extraction. The early layers capture low-level features like intensity gradients and boundary sharpness which are needed for outlining contours of the battery particles. Deeper filters aggregate simple edges into complex representation of crack network topology and fragmentation density.

Through testing, I found that fine-tuning only the classification head and final block gave me the best result. This means that the mid-level filters already contained enough texture recognition capabilities for SEM images. On the other hand, adding more filters or deeper layers did not improve performance possible because of the small 60 image sample size. Increasing the complexity beyond a certain point led to the model memorizing specific image features rather than learning generalizable degradation patterns.

OVERALL ANALYSIS, OBSERVATIONS AND INFERENCES FROM ALL TASKS

12. Why do you the three different tasks (Task A - regression, Task B - anomaly detection and Task C image classification) required very different performance metrics to assess their goodness? (**Word Limit: 100-150 words**)

ANSWER:

These tasks require different metrics as they address different problems and operational risks. For Task A, R^2 and RMSE quantify the precision of continuous ESI predictions, which is important for calculating exact battery lifespan. In task B, the dataset is highly imbalanced, 100 normal and 20 anomalies. Standard accuracy would be misleadingly high even if the model missed every anomaly. So AUC-ROC is used to assess the model's ability to distinguish between classes regardless of the threshold.

Finally, Task C was classification which utilizes Macro-averaged accuracy to ensure the model performs equally well across healthy, cracked, and fragmented categories. By choosing metrics that suit the failure costs, we ensure the ML models remain reliable and practically useful in an engineering environment.

13. Compare the three tasks (regression, anomaly detection, image classification). Which was hardest and why? Which required the most engineering intuition? (**Word Limit: 100-150 words**)

ANSWER:

I found Task C to be the most challenging as it required a shift from tabular data processing to spatial feature extraction. Initially, I struggled to engineer features from raw pixel intensities to represent crack geometry. I eventually realized that leveraging Transfer Learning with architectures such as ResNet was more effective, as these models come with pre-trained filters capable of identifying complex textures.

However, I felt Task B required the most engineering intuition. While Task C relied on automated feature extraction, Task B's performance was significantly enhanced by understanding physical sensor relationships. Identifying that a specific combination of pressure and flow signals indicated a pump anomaly needed a physics mindset.

14. What is one limitation of ML that you discovered through this project? And what is one engineering insight (not ML insight) you gained from working on these datasets? (**Word Limit: 100-150 words**)

ANSWER:

A limitation of ML is the model brittleness, which is when the algorithms fail when encountering data outside their training distribution. While the Task B model showed high AUC, its reliance on specific statistical baselines means that it might struggle with concept drift if deployed on a different pump model with different operational signatures.

The engineering insight I gained is that `pressure_ratio` is a more powerful indicator of pump health rather than individual raw sensor values. This is because the ratio captures the polytropic efficiency of the system, a physics principle that standard ML models might take a longer time to learn on its own. Providing this engineered feature allowed for a simpler or more robust model proving that physics informed inputs are indispensable for reliable engineering data analytics.

SELF-REFLECTION AND FEEDBACK

15. In your honest opinion, do you agree or disagree that EPD students are better positioned in applying ML techniques for engineering data analytics as compared to those from ISTD and ESD, even though they take specific data science / ML / DL / statistics courses in their core curriculum? (**Word Limit: 100-150 words**)

ANSWER:

I agree that EPD students hold a meaningful advantage over ISTD or ESD peers when applying ML to engineering data, though not because of deeper ML expertise. Our competitive edge lies in physics intuition and domain-specific knowledge. For instance, in Task B, someone without the relevant domain knowledge might rely on automated feature selection or polynomial expansions to find patterns. In contrast, an EPD student can leverage fluid mechanics to recognize that a pressure_ratio is physically significant indicator of pump health. Thus, EPD students should focus on leveraging domain knowledge as their primary advantage, treating ML as a sophisticated tool that complements, rather than replaces fundamental engineering reasoning.

16. In your honest opinion, how do you see yourself using these ML methods in the future EPD courses and where ? Are you confident that you can *self-learn* more advanced ML methods with the assistance of LLM / Gen-AI tools and use them pro-actively for solving complex engineering problems by yourself without assistance or with minimal assistance? Do you think a full semester module on machine learning or deep learning is necessary for *EPstuDents* to be able to apply them to solving real-world problems? (**Word Limit: 100-150 words**)

ANSWER:

I see myself using these ML methods in future EPD courses involving systems modelling and predictive maintenance, particularly in sensor-rich mechanical or
I can see myself using these ML methods in systems modelling and predictive maintenance within future EPD projects, particularly in sensor-rich mechanical or electromechanical systems.
The XGBoost regression pipeline from Task A is reusable for any tabular engineering dataset, and the CNN finetuning approach from Task C applies broadly to microscopy-based materials characteristics.
I can be confident that I can self-learn more advanced methods such as physics informed neural networks or graph neural networks for materials science with LLM assistance, as this project demonstrated that LLMs can effectively scaffold implementation while I contribute to domain reasoning.

17. In your honest opinion, how would you rate the experience of this 1D design project on a scale of 1 to 10 (10 being most satisfied and 1 being least satisfied)? What suggestions do you have to further enhance the offering of this design project for the next batch of students in terms of data sets, analysis methods, learning pedagogy, assessment process and submission timelines? *Your frank opinions are most welcome!* (**Word Limit: 100-150 words**)

ANSWER:

I would rate this as 5 out of 10. There are too many things that are not taught to us in this design project. ML algorithms or methods are abstract, and we do not know what most of them do. It could be self-learned, however there isn't enough time to learn it and I find it so confusing on how to apply what. It is dependent on generative AI/LLMs to complete the task, and it is basically a competition on how good the generative AI we used is.

On the other hand, when I was starting the project, I was interested in learning the new ML algorithms and neural networks and this project got me to utilize the models and gave me exposure to the different datasets and how they are trained.

